Physics Practice Worksheet

Topics: Kinematics and One-Dimensional Motion

Topic 2.1 & 2.2: Displacement, Velocity, Acceleration & Motion Graphs

2. Light travels at an approximate speed of 3×10^8 m/s.

a) Calculate the distance in miles that a pulse of light travels in 0.1 seconds (which is roughly the duration of a blink).

b) Compare this calculated distance to the diameter of the Earth.

3. A driver travels between cities using different constant speeds for various legs of the trip. She drives for 30.0 minutes at 80 km/h, 12.0 minutes at 100 km/h, and 45.0 minutes at 40.0 km/h. She also spends 15.0 minutes taking a lunch and gas break.

a) Determine the average speed for the entire trip.

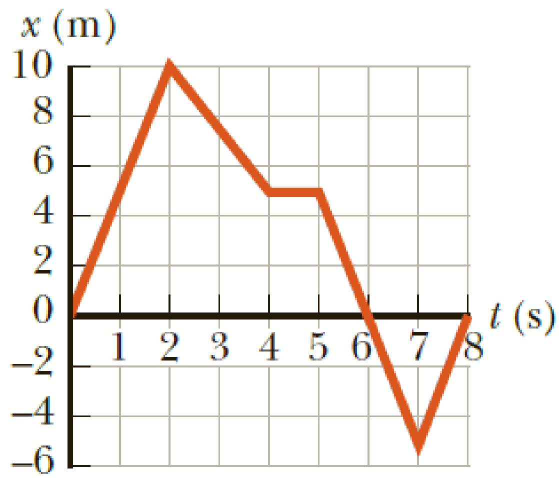
b) Determine the total distance travelled between the initial and final cities.

5. Two boats race across a 60-km-wide lake and back, starting at the exact same time. Boat A travels across at 60 km/h and returns at 60 km/h. Boat B travels across at 30 km/h, but upon realizing it is falling behind, returns at 90 km/h. Assuming turnaround times are negligible:

a) Determine which boat wins the race and by how much time (or state if it is a tie).

b) What is the average velocity of the winning boat for the round trip?

6. A graph of position versus time for a certain particle moving along the x-axis is shown below.



Based on this graph, find the average velocity in the time intervals from:

(a) 0 to 2.00 s, (b) 0 to 4.00 s, (c) 2.00 s to 4.00 s, (d) 4.00 s to 7.00 s, and (e) 0 to 8.00 s.

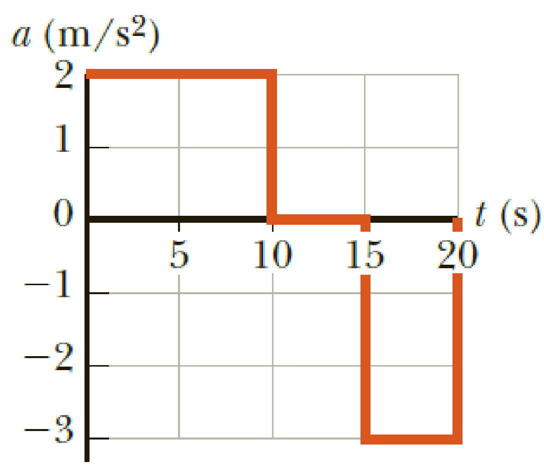
7. A motorist travels north at 85.0 km/h for 35.0 minutes and then takes a 15.0-minute rest stop. He then resumes driving north, covering 130 km in 2.00 hours. Calculate:
(a) his total displacement, and
(b) his average velocity for the entire trip.

11. A cheetah is capable of reaching a top speed of 114 km/h (71 mi/h). While sprinting after prey, a cheetah starts from rest and runs 45 m in a straight line, achieving a final speed of 72 km/h.
(a) Calculate the cheetah's average acceleration during this sprint.
(b) Find its total displacement at t=3.5 s.

13. A driver embarks on a trip, maintaining a constant speed of 89.5 km/h, except for a single 22.0-minute rest stop. If the overall average speed for the trip is 77.8 km/h, determine:
(a) the total time spent on the trip, and
(b) the total distance traveled.

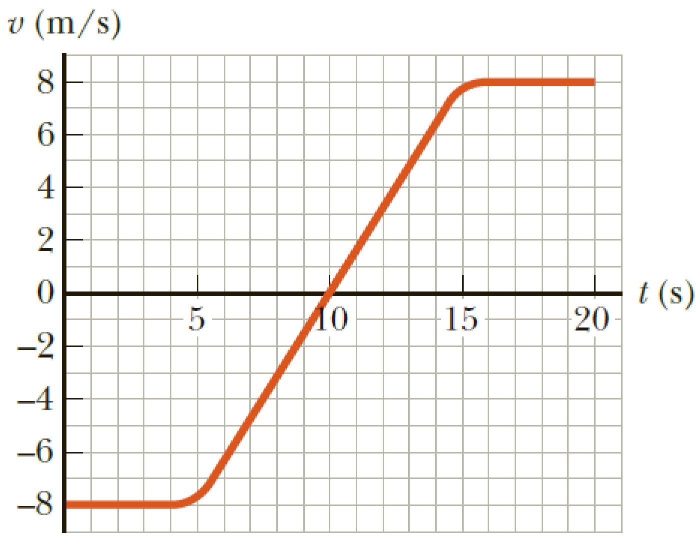
19. Runner A starts 4.0 miles west of a flagpole and runs due east at a constant velocity of 6.0 mi/h. Runner B starts 3.0 miles east of the flagpole and runs due west at a constant velocity of 5.0 mi/h. At the moment the two runners meet, how far are they from the flagpole?

20. A particle starts from rest and accelerates as shown in the figure below.



Determine
(a) the particle’s speed at t = 10.0s and at t = 20.0 s, and
(b) the distance travelled in the first 20.0 s.

24. The velocity vs. time graph for an object moving along a straight path is shown the figure below.



a) Find the average acceleration of the object during the time intervals
   i. 0 to 5.0s
   ii. 5.0 s to 15 s
   iii. 0 to 20 s
b) Find the instantaneous acceleration at
   i. 2.0 s,
   ii. 10 s,
   iii. (c) 18 s.

Topic 2.3: One-Dimensional Motion with Constant Acceleration

27. An object moving with a uniform acceleration has a velocity of 12.0 cm/s in the positive x-direction at the instant its x-coordinate is 3.00 cm. If its x-coordinate 2.00 s later is -5.00 cm, calculate its acceleration.

29. While uniformly slowing down to a final velocity of 2.80 m/s, a truck covers a distance of 40.0 m in 8.50 s. Find:
(a) the truck's initial speed,

32. An object moves with a constant acceleration of 4.00 m/s^2 and reaches a final velocity of 12.0 m/s over a certain time interval.
(a) If its initial velocity was 6.00 m/s, what is its displacement during this time interval?
(b) What is the total distance travelled during this interval?
(c) If its initial velocity was -6.00 m/s, what is its displacement during this interval?
(d) What is the total distance travelled during the interval described in part (c)?

33. During a test run, a car accelerates uniformly from rest to 24.0 m/s in 2.95 s.
(a) Determine the magnitude of the car's acceleration.
(b) How much time is required for the car to change its speed from 10.0 m/s to 20.0 m/s?
(c) Conceptually, will doubling the time interval always result in doubling the change in speed? Explain your reasoning.

34. A jet plane touches down on a runway with a speed of 100 m/s and can apply a maximum deceleration of -5.00 m/s^2 to come to rest.
(a) What is the minimum time needed for the plane to stop after touching the runway?
(b) Can this plane safely land on a small island runway that is 0.800 km long?

37. A train is moving along a straight track at 20 m/s when the brakes are applied, producing an acceleration of -1.0 m/s^2 as long as the train is in motion. How far does the train travel during a 40-s time interval that begins the exact moment the brakes are applied?

38. A car accelerates uniformly from rest, reaching a speed of 40.0 mi/h in 12.0 s. Calculate:
(a) the distance the car travels during this 12.0 s interval, and
(b) the constant acceleration of the car.

39. A car starts from rest and travels with a uniform acceleration of +1.5 m/s^2 for 5.0 s. The driver then applies the brakes, resulting in a uniform acceleration of -2.0 m/s^2. If the brakes are applied for 3.0 s, determine:
(a) how fast the car is moving at the end of the 3.0 s braking period, and
(b) the total distance the car has travelled from its starting point.

Topic 2.4: Freely Falling Objects

45. A ball is thrown vertically upward with an initial speed of 25.0 m/s.
(a) Determine the maximum height the ball reaches.
(b) How much time does it take to reach this highest point?
(c) How long does the ball take to hit the ground after reaching its peak?
(d) What is the ball's velocity at the instant it returns to its starting level?

46. From a height of 30.0 m, a ball is thrown directly downward with an initial speed of 8.00 m/s. How much time elapses before the ball strikes the ground?

47. An object is released from rest and falls freely. It takes 1.50 s to travel the final 30.0 m before hitting the ground.
(a) Calculate the velocity of the object when it is exactly 30.0 m above the ground.
(b) Determine the total distance the object travelled during its entire fall.

51. A tennis player tosses a ball straight up and catches it 2.00 s later at the exact same height from which it was released. Find:
(a) the acceleration of the ball while it is in flight,
(b) the velocity of the ball at its maximum height,
(c) the initial release velocity of the ball, and
(d) the maximum height the ball reaches.

53. A model rocket is launched straight upward with an initial speed of 50.0 m/s. It maintains a constant upward acceleration of 2.00 m/s^2 until its engines cut off at an altitude of 150 m.
(a) Describe the motion of the rocket after the engines stop.
(b) Calculate the maximum height reached by the rocket.
(c) How much time elapses after lift-off before the rocket reaches its maximum height?
(d) What is the total time the rocket spends in the air before hitting the ground?

54. A baseball is struck by a bat and travels straight upward. An observer notes that the ball takes 3.00 s to reach its maximum height. Calculate:
(a) the ball's initial velocity just after being hit, and
(b) the maximum height it reaches.

63. A ball is thrown upward from the ground with an initial speed of 25 m/s; at the same instant, another ball is dropped from a building 15 m high. After how long will the balls be at the same height?

65. A ball thrown straight up into the air is found to be moving at 1.50 m/s after rising 2.00 m above its release point. Find the ball’s initial speed.